VoiceServe — Progress Report

Nov 5, 2026 · @ARRAY

We have finished 3 of our 7 milestones: the base setup, a working practice student portal, and the first voice feature. A student can now ask "What's my attendance?" by voice or by typing, and gets a spoken answer and an on-screen card.

What VoiceServe is

VoiceServe is a voice assistant inside a university student portal. A student taps a mic button on any page and asks about attendance, fees, timetable or requests in plain speech. It answers, and it only makes changes (like filing a complaint) after the student says yes.

We do not touch the real university system. We built our own practice portal, called **Univra**, filled with made-up students. It copies the real portal's layout and menus but has its own name, logo and colours.

Progress at a glance

Three milestones are done. M4 is next.

#	Milestone	In simple words	Status
M1	Foundation	Project set up so everything starts with one command	Done
M2	Practice portal (Univra)	A working student portal with fake students	Done
M3	First voice feature	Ask about attendance by voice or text	Done (test mode)
M4	Full assistant	All features: fees, timetable, complaints, hostel leave, page navigation	Next
M5	Languages + rulebook	Hindi, Punjabi, Hinglish; answers from university rules with sources	Not started
M6	Staff side	Staff dashboard; hand over to a human when the student is upset	Not started
M7	Testing + report	Measure accuracy and speed, write the final evaluation	Not started

We also put up a public demo link (see "How to see the demo").

What we have built

M1 — Foundation

- One command (`make up`) starts all 9 parts of the system: the portal, the database, the voice server and the rest.
- Automatic checks run on every code change (GitHub Actions), including a scan that blocks passwords or keys from being saved in the code.
- Every design decision is written down with its reason (40 decisions so far).

M2 — The Univra practice portal

- **Login** with registration number and password. The account locks after 5 wrong tries.
- **Portal layout like the real UMS:** announcement strip, top menus (Academics, Administrative, Important Links, Student Services, Quick Links), search, side bar, student name and section.
- **Dashboard:** shortcut buttons, news, teacher messages, my courses with attendance circles, fee due, pending assignments, events, and a pop-up for placement drives.
- **RMS (complaints):** the student agrees to 5 declarations, fills the form, gets a request ID, and can see history and give feedback. If a similar complaint is already open, it suggests adding to that one instead of making a duplicate.
- **Fake data:** about 600 made-up students with realistic attendance, marks, fees and complaints. Six fixed demo students each show a special case: low attendance, overdue fee, a locked account, and so on.

M3 — First voice feature

- A mic button on every page. The student holds it and asks about attendance ("Am I short in Computer Networks?").
- The assistant answers out loud and shows a card with the numbers. All numbers come from the database, never guessed by AI.
- The student can interrupt it while it speaks.
- Typing works too, if the mic is not available.
- We measure how long each answer takes.

How we keep it safe

These rules are built into the code, not just written in instructions to the AI.

- **Fake data only.** No real student data, no real university system. Fake phone numbers start with 50, which no Indian mobile number does.
- **No looking at other students' data.** The system only uses the logged-in student's identity. Asking for someone else's data is refused.
- **No made-up numbers.** Attendance %, fees and dates come straight from the database. The AI never invents them.
- **No double complaints.** If a request is sent twice by mistake, only one is saved.
- **Easy-to-read request IDs.** Each ID has a check digit, so a misheard or mistyped number is caught.
- **Private logs.** Registration numbers and phone numbers are hidden in log files (for example `****78`).
- **Spending cap.** Paid AI services stop at $5 a day.

How we checked it works

Check	Result
Automatic code tests	259 passing
Browser tests (a robot clicks through the portal)	13 passing
Voice test with a recorded question	Passes: question heard, answer spoken, card shown, interruption works
Time from end of question to first spoken word	0.62 seconds (target: under 1 second)
Fake students below 75% attendance	About 26% (realistic mix, as planned)

The voice numbers are from **test mode**, which uses recorded speech instead of paid speech services. The real speech service will add some time; we will measure it once the service keys are set up.

How to see the demo

1. On the team laptop we run `make up`, then open the portal in a browser.
2. For a link anyone can open, we run `make share`. It gives a temporary public web address that works while the laptop is on.
3. Log in as a demo student (accounts are shown on the login page), open the dashboard and RMS, then press the mic and ask: "What's my attendance?"

A good path to show: log in as Aarav (one course below 75%), ask about attendance, then log a hostel complaint in RMS and see it in RMS History.

Challenges and how we solved them

Problem	What we did
No access to real university data	Built our own portal with fake students that copies the real layout
Must not copy the real university's branding	New name (Univra), logo and colours; only the structure and menus match
Free hosting for the demo did not work out (the free plan didn't cover it)	Share the demo from the laptop through a free secure tunnel (`make share`)
Voice calls need special network ports	Use LiveKit Cloud's free plan for the audio connection
Speech recognition missed Indian English and portal words	Tuned it for Indian English and added words like "RMS" and course codes
Paid AI services cost money	Test mode with recorded speech, plus a $5 daily spending cap

Next steps

M4 (next): the assistant will answer about fees, timetable, exams and assignments. It will file complaints and hostel leave after reading them back and getting a clear "yes", and it will open any portal page on request.

After that: Hindi, Punjabi and Hinglish (M5), the staff dashboard (M6), and the full testing report (M7).

Questions for you

- [ ] Is it fine to use paid speech and AI services (about $5 a day at most) for the real voice tests?
- [ ] Should phone-call access stay an optional extra, or is it required for the final demo?
- [ ] Can we record volunteer students speaking, to test Hindi and Punjabi accuracy? Do we need a consent form?